

CONCEPT NOTE

VaaniAI — AI-Powered Voice Agent for e-Governance

Government of Rajasthan

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1. EXECUTIVE SUMMARY

VaaniAI proposes an AI-powered multilingual voice agent platform designed to transform citizen engagement for the Government of Rajasthan. The solution enables 24/7 automated voice-based interactions in Hindi, English, and Rajasthani dialects, handling citizen queries about government schemes, grievance registration, appointment booking, and service status tracking.

By leveraging cutting-edge AI technologies including Natural Language Processing (NLP), Automatic Speech Recognition (ASR), and Text-to-Speech (TTS), VaaniAI eliminates long wait times, reduces operational costs, and ensures consistent, accurate information delivery to citizens across the state.

2. PROBLEM STATEMENT

The current citizen service delivery mechanism faces several critical challenges:

- High call volumes at government helplines leading to long wait times (average 15-20 minutes)
- Limited operating hours (typically 9 AM to 5 PM) restricting citizen access
- Language barriers — many citizens are more comfortable in Hindi or Rajasthani dialects
- Inconsistent information delivery due to varying agent knowledge levels
- High operational costs for maintaining large call center teams
- Lack of data-driven insights into citizen needs and service gaps
- Manual grievance tracking leading to delayed resolution and poor follow-up

3. PROPOSED SOLUTION

VaaniAI provides an end-to-end AI voice agent platform with the following core capabilities:

3.1 AI Voice Agent

- Real-time speech recognition and natural language understanding in Hindi, English, and Rajasthani
- Context-aware conversational AI with memory for multi-turn dialogues
- Dynamic knowledge base integration for up-to-date scheme information
- Seamless handoff to human agents for complex queries

3.2 Citizen Service Features

- Scheme Information: Detailed information about 100+ government schemes with eligibility checking
- Grievance Registration: Automated complaint filing with reference number generation
- Appointment Booking: Schedule visits to government offices
- Application Status Tracking: Real-time status updates on pending applications
- Document Guidance: Step-by-step guidance on required documents for services

3.3 Administrative Dashboard

- Real-time analytics on call volumes, resolution rates, and citizen satisfaction
- Automated reporting for departmental performance monitoring
- Knowledge base management for easy content updates

- Campaign management for proactive citizen outreach

4. TECHNOLOGY ARCHITECTURE

VaaniAI is built on a cloud-native, microservices architecture leveraging enterprise-grade Azure services:

Component	Technology	Purpose
Speech Recognition	Azure AI Speech	Real-time ASR in multiple languages
NLP Engine	Azure OpenAI GPT-4	Intent recognition & response generation
Text-to-Speech	Azure AI Speech	Natural voice synthesis in Hindi/English
Telephony	Smartflo / SIP Trunk	Call routing and IVR integration
Database	Cloud Database	Citizen data and interaction logs
Analytics	Custom Dashboard	Real-time monitoring and reporting
Security	Azure Security Center	End-to-end encryption & compliance

5. EXPECTED IMPACT & MEASURABLE OUTCOMES

The deployment of VaaniAI is expected to deliver significant improvements across multiple dimensions:

Metric	Current State	Expected Outcome	Improvement
Call Handling Time	15-20 mins avg	1-2 mins avg	90% Reduction
Citizen Satisfaction	~40%	~95%	70% Improvement
Grievance Resolution	~50% success	~90% success	80% Increase
Citizen Engagement	Limited hours	24/7 availability	50% Rise
Operational Cost	High (manual)	Automated	30% Savings
Language Coverage	Mostly English	Hindi/English/Rajasthani	3x Coverage

6. SCALABILITY & SUSTAINABILITY

VaaniAI is designed for phased deployment ensuring minimal risk and maximum adoption:

Phase 1 — Pilot (3 months):

- Deploy in 1-2 departments (e.g., Jan Soochna Portal, CM Helpline)
- Handle up to 10,000 calls/day with 5 concurrent channels

Phase 2 — State-wide Rollout (6-12 months):

- Expand to all major departments across Rajasthan
- Scale to 100,000+ calls/day with 50+ concurrent channels

Phase 3 — Multi-State / National (12-24 months):

- Replicate the model across other Indian states
- Add regional languages (Tamil, Telugu, Bengali, etc.)
- Integration with national platforms (UMANG, DigiLocker)

7. SUPPORT REQUIRED FROM DEPARTMENT

- **Data Access:** Secure API access to citizen databases, scheme repositories, and grievance management systems for AI model training and real-time operations.
- **Policy Support:** Official adoption mandate, legal frameworks for AI-based citizen interactions, and data governance policies aligned with IT Act provisions.
- **Infrastructure Support:** Integration with existing IT systems (e-Mitra, SSOID, Jan Aadhaar), dedicated DID numbers, and network connectivity at government offices.
- **Stakeholder Coordination:** Designated nodal officers for each department to facilitate knowledge transfer and content validation.
- **Feedback Mechanism:** Citizen feedback channels to continuously improve AI accuracy and service quality.

8. INDICATIVE BUDGET ESTIMATE

Component	Pilot Phase	State-wide Phase	Remarks
Platform Setup	₹ 5,00,000	—	One-time setup
AI Services (Monthly)	₹ 2,50,000	₹ 15,00,000	Azure AI consumption
Telephony (Monthly)	₹ 1,50,000	₹ 8,00,000	DID & call charges
Maintenance (Monthly)	₹ 1,00,000	₹ 5,00,000	Support & updates
Training & Onboarding	₹ 2,00,000	₹ 5,00,000	Staff training

Note: Actual costs may vary based on call volumes, number of departments, and specific requirements. A detailed commercial proposal will be provided upon request.

9. IMPLEMENTATION TIMELINE

Phase	Duration	Key Activities
Discovery & Planning	Weeks 1-2	Requirements gathering, stakeholder meetings
Development & Integration	Weeks 3-6	AI model training, system integration
Testing & UAT	Weeks 7-8	User acceptance testing, performance tuning
Pilot Launch	Weeks 9-10	Go-live in pilot department(s)
Optimization	Weeks 11-12	Performance monitoring, feedback incorporation
State-wide Rollout	Month 4-12	Phased expansion across departments

10. CONCLUSION

VaaniAI represents a transformative opportunity for the Government of Rajasthan to modernize citizen service delivery through AI-powered voice technology. By automating routine interactions, providing 24/7 multilingual support, and delivering data-driven insights, VaaniAI will significantly enhance transparency, efficiency, and citizen satisfaction.

We look forward to partnering with the Department of IT & Communications to bring this vision to life and set a benchmark for AI-driven e-Governance in India.

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